

SCIENCE / PHILOSOPHY / IDEAS

THE WORLD DOES NOT READ EQUATIONS

What the laws of nature are,
where they come from
and how much we can understand

Order, emergence and the limits of understanding.
The possibility of a simulated world.

2026

Contents

Prologue. The morning it falls differently 4

How to read this book..... 6

I · HOW THE WORLD CAME TO HAVE LAWS

01. A regularity is not yet a law 7

02. When nature had no legislator 9

03. God, geometry and the world machine 12

04. Hume and Kant: the rift between world and mind..... 14

05. Do laws govern or summarize? 17

II · HOW A WORLD’S REGULARITIES ARISE

06. An atlas of laws 20

07. Symmetry: what remains when we change our viewpoint..... 23

08. The arrow of time and order born of probability 25

09. How new rules arise without a new legislator 27

10. More than the sum of particles..... 30

11. Life: regularities with memory..... 33

12. Can laws change?..... 36

Contents · continued

III · THE MIND TRYING TO UNDERSTAND

13. The brain that demands explanations.....	39
14. What does understanding actually mean?	42
15. Are there things even intelligence cannot compute?	45
16. Faith, meaning and alternatives to mechanism.....	48

IV · WHO BENEFITS WHEN NATURE SEEMS TO HAVE DECIDED

17. Who speaks in nature's name	51
18. Profit, merit and the supposed laws of the market	54
19. Art between constraint and invention	57

V · THE UNIVERSE AS A PROGRAM: POSSIBILITY, EVIDENCE, RESPONSIBILITY

20. The simulation hypothesis without cinema	59
21. What would count as evidence?	62
22. A protocol for a suspicious world	65
23. If the rules are rewritten tomorrow.....	68
24. How to live without the final explanation.....	71

TOOLS · FOR READERS WHO WANT TO CHECK

A. Four tools for not confusing story with law.....	73
B. Glossary: words that change the question.....	76
Bibliography	79

Prologue. The morning it falls differently

Imagine that tomorrow morning a laboratory announces something almost ridiculous: two highly precise clocks operating through different mechanisms no longer maintain the same ratio between their rates. The team checks temperature, power supply and analysis software. Then it replaces the instruments. The difference remains. Other laboratories reproduce the observation. It is not a fluke, but nobody yet knows what it is.

One news story will say the laws of nature have changed. Another, that the universe has received an update. A cautious physicist will say something has changed in the relationship between measurements and model. The three sentences do not carry the same weight. The first interprets; the second, for now, invents a cause; the third defines a problem.

The scene is imaginary. Its question is real: what would have to happen for the expression ‘law of nature’ to lose its meaning? A single deviation? A century of different regularities? The discovery of deeper rules? Or the discovery of a programmer?

We are accustomed to saying that the universe obeys laws. The verb hides a small drama. Obeying a human law presupposes that violating it is possible. A pedestrian can cross against a red light. An electron does not seem to choose between obedience and revolt. When a result contradicts a physical law, we usually do not put the electron on trial: we check the law, conditions and measurement.

Yet the idea of law does something extraordinary. It allows us to use what has happened to anticipate what has not happened yet. It transforms experience into a transmissible resource. At the same time, it can turn a habit into destiny: ‘That is human nature’, ‘That is how the market works’, ‘The weak always lose’. Sometimes we know a constraint. At other times we hide behind it.

This book follows both powers of the concept. The first builds tools, explanations and possibilities for intervention. The second closes discussions before they begin. We have no reason to confuse them or to give up the first because we fear the second.

We will also reach a possibility seemingly taken from cinema: a simulated world capable of discovering that it is simulated. The interesting problem is not whether we can invent such a story. We can invent countless ones. The problem is whether the story bears the risk of being wrong and whether observations exist that separate it from competing explanations.

Our starting position is deliberately modest: the world can contradict us. It does not follow that we know what its laws ultimately are. It does follow, however, that the difference between a good explanation and a convenient one matters. Perhaps the universe contains no sentences. But some of our sentences survive encounters with it much better than others.

How to read this book

Parenthetical citations refer to the bibliography at the end. Philosophical texts are presented through interpretations and paraphrases, not translated quotations claiming a philological precision that is absent. Examples involving imaginary laboratories, communities, programs and observers are thought experiments constructed for this book. They do not represent empirical results.

Three levels will recur: what is well supported experimentally; what is a disputed philosophical interpretation; and what is a speculative hypothesis from which we can demand testable consequences. They are not three rungs of the same ladder. A mathematical proof is rigorous relative to its premises; an experiment connects a model to part of the world; a metaphysics asks what kind of world could make such connections possible.

‘Natural law’ can also designate a moral or legal doctrine. Here, ‘law of nature’ will primarily designate regularities and descriptive principles about the world. The transition from these to moral obligations will be discussed separately. That something happens is not, by itself, an argument that it ought to happen.

01. A regularity is not yet a law

All the keys on my table are cold. If I declared this finding ‘the first law of living-room metals’, there would be an immediate argument against it: I can heat one of the keys. But the problem runs deeper. Even if nobody ever heated them, the sentence would remain an accidental description of a collection, not an explanation of how metals behave.

A law claims more than fidelity to an inventory. It concerns stable relationships and what would happen in situations different from actual ones: counterfactuals. This ‘if’ connects law to possibilities, not merely completed facts. The exact relationship between laws and counterfactuals remains disputed (Carroll, 2020).

Take a spring. Within a certain deformation range, we can construct a model in which restoring force is proportional to extension. The model does not say that every elastic object behaves this way under all circumstances. Stretch the spring too far and the relationship ceases to be useful. We discover a domain limit, not nature’s decision to become unlawful.

This yields a practical question worth asking alongside every law taught in school: under what conditions can it be used? A formula without a domain resembles a medicine without indications and contraindications. It can be correct in form and misused in the world.

Four things must also be separated. An observation is a situated result: the instrument recorded a value. A regularity is a pattern among results. A law is a relationship to which we attribute a certain stability and generality. A theory is a structure of concepts, assumptions and relationships organizing explanations and predictions. A theory does not turn into a law after accumulating enough confirmations, like an employee being promoted. The two words serve different functions.

A model is a selective representation used for a problem. It may contain laws, approximations and conventions. A subway map is not false because it fails to preserve all distances; it is useful because it elimin-

ates details unsuited to its purpose. A map for evacuating a building must preserve different relationships, however. Evaluating a model also means asking what information it must conserve.

Here the first ambiguity of realism appears. Are the letters in the formula real? They are signs produced by humans. Is the described relationship real? It may be: two teams can use different notation and build devices that work the same way. Is there, additionally, an entity called a 'law' compelling things to behave that way? This is a further metaphysical thesis. The formula's success does not automatically decide it.

Confusing these levels produces two exaggerations. The first says: because equations are invented, the world they describe is invented. But a house's address is also a convention, without making its walls conventional. The second says: because some equations work remarkably well, they are precisely the language in which the universe was written. This conclusion is not mandatory either. Multiple representations can capture the same structure without any being reality's ultimate alphabet.

It is reasonable to consider a law good when it makes precise predictions, works in new situations, connects previously separate phenomena and allows us to identify when it can no longer be used. No criterion is sufficient alone. A very vague statement can 'explain' everything precisely because it excludes nothing. A relationship fitted impeccably to known data may fail at the first new observation.

The decisive question becomes: what risk does the explanation take? If every possible result can be reinterpreted as confirmation, we may have a worldview but not yet a tool for discriminating among possible worlds. A law's power does not lie in the word's solemnity. It also lies in the number of things it allows us to declare incompatible with it.

02. When nature had no legislator

A person can anticipate seasons without believing there are ‘laws of nature’ in the modern sense. They can build, cultivate and navigate using practical regularities. The history of this concept must therefore not be confused with the history of intelligence. The absence of our vocabulary does not mean an absence of observation, reasoning or technology.

In Physics, Aristotle distinguishes natural things by their having within themselves a principle of motion and rest. A bed is made by an art; a tree develops through its nature. Aristotelian explanation pursues several kinds of ‘why’: material, form, source of change and purpose. These are the four causes, and ‘cause’ here has a broader meaning than the blow setting a ball in motion (Aristotel, Fizica, II).

This is not a clumsy version of Newtonian mechanics. It is a different distribution of questions. Why does an organism grow in a particular way? What is it made of? What structure characterizes it? What is a part’s function? Contemporary science does not answer all these questions through a single type of explanation, even though it does not adopt Aristotelian physics.

It would be equally wrong to treat Chinese thought as Western physics lacking equations. In Dao De Jing, order is conceived through Dao, relationships, transformations and ways of acting that do not force things. Chapter 25 culminates, in common translations, in the idea that Dao follows what is so of itself, associated with the term *ziran*. Here we have neither a quantifiable instruction code nor a demonstrated prototype of systems theory. We have a conceptual alternative: order can be imagined without the metaphor of a sovereign issuing commands (Laozi, Dao De Jing, cap. 1, 25, 37).

The comparison is fruitful if we preserve the difference. We can ask what we gain by conceiving order as spontaneous regularity and what we lose by trying to translate all of nature into obedience. We cannot

announce that an ancient text discovered quantum mechanics merely because it uses suggestive words about transformation and relationship.

Another challenge appears in Nāgārjuna's Indian philosophy. His critique of existence through an independent intrinsic nature, *svabhāva*, emphasizes dependencies and the difficulty of isolating an absolute foundation. This is not the claim that nothing works or every outcome is possible. The analysis concerns how concepts and things are dependent within a specific philosophical and liberating project. It is not an experimental anticipation of relational physics (Westerhoff, 2026).

It is nevertheless worth using as a question directed at our own reflexes. When we say an electron 'has a nature', do we mean an isolated essence or its behavior in a network of interactions? When we call an object 'independent', independent of what exactly? Philosophical comparison can open questions without retroactively granting scientific certificates.

In theological traditions, order and necessity have likewise been separated. Debates about al-Ghazālī show how risky the summary 'religion denied causality' is. A central question concerned whether the connection between events is necessary in itself or depends on God. Scholars' interpretations differ regarding the relationship among occasionalism, secondary causes and his other writings. Challenging the connection's metaphysical necessity does not simply amount to denying observable sequences (Griffel, „Al-Ghazali“).

These traditions do not offer the same doctrine in different costumes. Aristotle, Laozi, Nāgārjuna and Islamic theologians discuss partly distinct problems in contexts with different aims. Mixing them into a 'universal wisdom' would be as reductive as declaring them mere errors preceding modern science.

The common lesson is more sober: the metaphor of law was not inevitable. We could have organized experience around powers, relationships, rhythms, forms or dependencies. That the language of laws has proved highly productive does not mean all other questions have become meaningless.

This history also has a critical advantage. When someone says ‘the human mind cannot think of nature otherwise’, we can reply: it already has. What remains to be established is not whether alternatives are imaginable, but what they can explain, what they allow us to measure and where they mislead us.

03. God, geometry and the world machine

A human law has an author. Where, then, does the law obeyed by planets come from? In a culture for which creation and divine order were central questions, the vocabulary of the cosmic legislator was not mere decoration. It offered a bridge between theology, metaphysics and the new mathematical physics.

For Descartes, matter is characterized by extension and its changes are explained mechanically. The stability of order is linked to God's unchanging character. In *Principles of Philosophy*, the laws of motion do not appear merely as practical summaries of observation. They have a justification within his philosophical architecture. This confidence did not prevent some Cartesian collision rules from being wrong. A metaphysical justification strong for its author does not guarantee good physics (Descartes, 1644, part II-a; Slowik, „Descartes' Physics“).

Newton radically changes the power of mathematical unification. *Principia* connects celestial and terrestrial motions through a common framework of principles and demonstrations. Its importance lies not merely in writing formulas but in showing how very different phenomena can be related through explicit assumptions, approximations and comparisons with observation (Newton, 1687/1726).

This unification contains an intellectual promise: we do not necessarily need a separate explanation for every region of the world. But the promise does not close every question. Successfully describing gravitational attraction does not, simply by writing the relationship, explain gravity's ultimate mechanism. Newton himself distinguishes what can be deduced from phenomena from hypotheses about causes that he cannot establish in the same way.

Imagine two astronomers. The first has an intuitive story about cosmic vortices but cannot obtain sufficiently good trajectories from it. The second has a precise mathematical relationship, yet the underlying mechanism troubles him. Whom do we follow? For navigation and

calculation, whoever predicts better. For the question about the nature of interaction, the dispute can continue. This is an important moment: predictive usefulness and metaphysical satisfaction can part ways.

The modern concept of law thus gains autonomy. We need not solve every question about the universe's author to compare two models of motion. Believer and nonbeliever can share a laboratory if they accept the same procedures of measurement and correction. This is a logical possibility of the method, not a claim that the actual history of science was free of religious and political conflicts.

The machine metaphor also changes what we consider explained. If understanding means showing the parts and their interactions, purpose risks being excluded from nature and relocated to the designer. The machine has a purpose for its builder, not necessarily for its gears. Such a model clarifies many phenomena but may become too narrow when used as a premature verdict on organism, mind or society.

A historical qualification is also necessary: 'science simply replaced religion' is too linear a story. For these authors, mathematical, theological and philosophical questions coexist. What gradually changes also includes where evidence is demanded. A claim about trajectories can be confronted with measurements, however venerable the metaphysics from which it began.

Who benefits from this shift? In principle, anyone who can use a reproducible result without belonging to the circle that formulated it. A public relationship can weaken an authority's monopoly over explanation. Who loses? The authority claiming that the speaker's status suffices to decide how nature behaves. But a new risk immediately appears: mathematical expertise can be used as authority in matters calculation does not resolve.

Secularizing the method therefore does not eliminate the priestly temptation. It can change its uniform. The relevant difference is not whether the speaker wears a robe or a lab coat, but whether they show how they can be contradicted, what they assumed and how far their result extends.

04. Hume and Kant: the rift between world and mind

Every day we light fires, press switches and cross bridges assuming known regularities will continue. We do not hold a seminar on induction before every gesture. David Hume nevertheless observes something uncomfortable: that a sequence has repeated does not entail, through pure logic, that it must repeat tomorrow (Hume, 1748, sec. IV–V).

The argument is not that everything will become chaotic tomorrow. It is that the justification ‘it has always worked’ already uses a transfer from past to future. If we justify the transfer by saying such transfers worked in the past, we return to what needed justification. Experience provides extraordinary practical support, but not a deduction making every other future impossible.

Hume thus separates logical certainty from expectation formed through experience. Habit plays a central role in how the mind connects events. I see the first event and expect the second. But where do I see necessity itself? In observation I encounter events and sequences, not a metaphysical force labeled ‘must’ (Hume, 1748, sec. VII).

This critique does not demolish research. It removes an overly convenient justification. We can seek more robust models, vary conditions, test in new situations and organize confidence by degrees. The philosophical problem of induction remains, but it does not make all forecasts equal. Lacking an absolute guarantee does not mean having no reason.

Imagine two people planning to build a bridge. One uses experiments, calculation and independent checks. The other rolls dice to choose dimensions. Hume does not make them epistemically equivalent. The first person’s inability to prove nature’s eternal uniformity does not erase their advantage relative to the goal. Skepticism becomes confused when it demands of the prudent person an infallibility the imprudent one does not even try to offer.

Kant responds by shifting the question: what conditions make experience of an objective world possible for us? We do not simply receive a ready-organized stream for the mind to photograph. In his view, experience presupposes forms and categories through which objects can be known; causality holds a central place. Knowledge of nature is linked to this contribution of the subject (Kant, 1783, §§14–38; 1781/1787).

This does not mean individual desire can decide an experiment's outcome. When Kant discusses the mind's contribution, he does not claim that each person arbitrarily invents gravity. He examines the general conditions of possible experience. The distinction between such conditions and a law's empirical content is essential: the structure of knowledge does not by itself supply every equation of physics.

We can formulate the tension through an example. A camera has a sensor selecting part of the light. The photograph depends on the world and the device. Noticing the device's role does not prove that the photograph shows nothing real. But neither does it allow us to confuse the photograph with the photographed thing in every respect. The analogy has limits: Kant discusses conditions far more fundamental than a technical device. Yet it shows why 'depends on the mind' and 'is arbitrary' are different propositions.

Later criticism may ask whether certain structures considered necessary are actually revisable. Conceptual progress sometimes forces us to learn ways of thinking that contradict everyday intuition. We need not declare the entire Kantian dispute settled to acknowledge this possibility. But developing a new physical theory does not automatically resolve the philosophical problem of our access to the world either.

Hume prevents us from confusing past success with absolute necessity. Kant prevents us from forgetting the mind's organizing contribution. Between them opens the space in which we still work: trying to know something independent of our desires with conceptual tools that are not independent of us.

It does not follow that truth is impossible. It follows that a mature picture of knowledge must be able to say two things simultaneously: there is resistance from the world and selection by the mind. Without

the first, any story would suffice. Without the second, we would not understand why discovering a law sometimes also means inventing a new way to ask the question.

05. Do laws govern or summarize?

Suppose we have before us the entire history of a universe: every event, without mistakes or gaps. Would anything remain to learn about its laws? The question separates several philosophies that can accept the same experiments and interpret differently what those experiments tell us exists.

The Humean tradition begins with the distribution of facts. In the best-system approach associated with David Lewis, laws are not additional commands issued to events. They belong to the system achieving a privileged balance among simplicity, informative power and, for probabilistic theories, fit with the facts. Not every convenient summary becomes a law (Carroll, 2020).

The advantage is ontological economy. We do not introduce an invisible administrator of necessity alongside events and properties. The difficulty is that simplicity sometimes seems vocabulary-dependent: what is complicated in one notation may be short in another. Another concern arises: if laws depend on the entirety of history, in what sense do they explain the very history they summarize? A defender of this position may reply that an explanation need not be a separate cause of the facts.

Armstrong proposes another direction. Laws are relations among general properties, called universals, not merely coincidences among particular cases. The relationship among universals would support the pattern's necessitating character. This conception tries to distinguish a genuine regularity from a merely repeated accident within the world's own structure (Armstrong, 1983).

The price is a new question: what makes the relation among universals compel particular instances? If we answer only 'it is a relation of necessity', have we offered an explanation or named the problem? The objection is not fatal by definition; every theory reaches elements it no longer explains through others. But where explanation stops must be made visible.

A third family emphasizes powers and dispositions. An object would not be a mere bearer of labels awaiting external commands; its properties would include ways of reacting. A glass's fragility refers to what would happen in certain circumstances even if the glass is never broken. Laws might express these powers. Within the broader debate, positions on dispositions are not identical and do not all imply the existence of separate fundamental laws (Carroll, 2020).

This picture has an intuitive merit: it places order within things. It also has a difficulty: if we identify a property only by what it produces and explain the production through the property, we risk circularity. To avoid mere wordplay, we must show independent relationships among manifestations, conditions and structures.

Maudlin defends treating fundamental laws as irreducible elements of ontology. Not every fundamental notion in physics must be reconstructible from an inventory of lawless facts. Some things may be primary in explanation. His position directly challenges the Humean reduction project (Maudlin, 2007).

The answer is intellectually respectable but unsatisfying to anyone always asking 'why?' Laws would be what explains, not what still needs explanation. Yet the opponent does not automatically escape a stopping point: they may leave the total distribution of facts unexplained. The dispute is partly about where we accept a basic fact.

Nancy Cartwright challenges the assumption that the success of laws in controlled arrangements automatically authorizes a uniform picture of the entire world. The metaphor of the 'dappled' world foregrounds capacities, local conditions and mechanisms producing regularities. The lesson for science is to investigate conditions for transporting a result, not assume a formula travels without baggage. This is a philosophical thesis about the power and limits of laws, not an experimental verdict that any unification is impossible (Cartwright, 1999).

Finally, we can be cautious about all these ontologies. Constructive empiricism associated with Bas van Fraassen demands empirical adequacy without requiring belief in the complete truth of the unobservable description. Structural realism, in different forms, places special

trust in relationships or structures preserved by theories even when the picture of objects changes. They are not the same program: one limits commitment; the other selects it (Chakravartty, 2017).

This book's working position is a critical realism of constraints. Some relationships resist us and permit stable interventions. This is a reason not to treat the world as arbitrary fiction. But from a constraint's robustness we do not automatically deduce that we have discovered a legislator, an ultimate necessity or the only possible description. We can have excellent reasons to use a law and insufficient reasons to declare its metaphysics settled.

06. An atlas of laws

The expression 'law of nature' shelters claims that do not all work the same way. If we judge them by a single template, we will declare some disciplines insufficiently scientific and grant others a precision they do not possess.

A dynamical law describes how a state evolves. We know the state at a given time, specify interactions and try to calculate what follows. In a deterministic model, complete initial data and the same rule yield the same evolution. The word 'complete' does much work. In practice, data are finite and the problem may be highly sensitive to errors.

A probabilistic law describes distributions of outcomes, not necessarily the individual outcome. If a model assigns an event a probability of one quarter, its occurrence does not refute the model, nor does its nonoccurrence in a single trial. Evaluation requires series, comparable conditions and procedures accounting for random variation. Claiming that every individual surprise violates a probabilistic law means demanding something other than what the model asserts.

A conservation principle says a quantity is preserved under relevant conditions. It is not necessarily a complete instruction for every movement. Knowing the total money in a system of accounts does not alone tell us who makes the next transfer. Conservation restricts possibilities without always choosing a single one. The accounting analogy explains a constraint's role; it does not turn energy into currency.

Constitutive laws describe the response of a particular material type: the relationship between stress and strain, or current and voltage under a particular regime. Some are excellent approximations. Their domain is not a hidden defect but part of their meaning. A material model can have parameters dependent on temperature and history without making every prediction impossible.

Then there are scaling regularities in which one quantity varies as a power of another. Such a fit may suggest a mechanism but does not identify it alone. Multiple processes can produce similar mathematical

forms. A straight line in a convenient representation begins an investigation; it does not prove that the same principle governs cities, organisms and markets.

Another type of principle formulates limits or impossibilities: which operations cannot be performed within a theory with particular resources and conditions. Such a prohibition can be stronger than a specific prediction. But the premises must be retained. 'It cannot be built in this model' does not automatically mean 'it cannot exist in any logically conceivable reality'.

These differences are supplemented by a separate axis: fundamental or effective. An effective law describes a domain of energies, scales or conditions without claiming to enumerate all ultimate mechanisms. Effective theories are not mere patches awaiting truth. They can systematically organize which effects matter and how large neglected corrections are (Burgess, 2007).

In research practice, 'fundamental' does not mean we possess the universe's final certificate. It may mean fundamental relative to the current framework or a level of explanation. We should be alert when a methodological adjective turns, without argument, into a metaphysical verdict.

Physical laws must also be distinguished from mathematical truths. A theorem shows what follows from axioms and definitions. To use it about the world, we must still justify the correspondence between the formal structure and the investigated system. We can prove a result perfectly about a model that describes its object poorly. Deductive rigor does not automatically repair the choice of premises.

No definition is an empirical discovery merely because it carries symbols. If we define a quantity as the ratio of two others, the relationship is true by convention; what needs investigation is how those quantities behave and why the definition helps. Sometimes confusion arises precisely when an accounting identity or definition is presented as a causal explanation.

This atlas offers a simple discipline. Before asking whether a law is true, we ask what kind of claim it is, what domain it has, what data would count against it and which part of the conclusion comes from

definitions. The word 'law' does not replace these questions. A science book becomes more useful by explaining differences than by collecting formulas under the same solemn label.

07. Symmetry: what remains when we change our viewpoint

Rotate a round table about its center. Its shape stays the same. This is a simple geometric symmetry. In physics, the idea can become much more powerful: we systematically change an experiment's description or arrangement and ask what remains unchanged in the theory's structure.

Not every change is a symmetry. If we move an experiment from one environment to another, we may have changed physical conditions, not merely its position on a map. To formulate an invariance, we must specify what we transform and what we preserve. A vague symmetry is as unhelpful as an equality in which we do not know what we compare.

Emmy Noether's theorems show, under precise conditions of the variational formulation, a link between continuous symmetries and conservation relations. In standard examples, time-translation invariance is associated with energy conservation, spatial-translation invariance with momentum, and rotational invariance with angular momentum. But there is no universal rule that 'any symmetry produces any conservation': the nature of the symmetry, action and problem conditions matter (Noether, 1918/1971).

The important intuition is this: if shifting the time origin does not change the dynamics' structure for a relevant closed system, the theory does not arbitrarily privilege one moment through a special instruction. This formal uniformity can yield constraints on evolution. A conservation principle can thus be linked to the theory's architecture rather than added as an isolated rule.

There is a popular trap. People sometimes pass directly from 'energy is conserved' to claims about the entire universe without checking whether total energy and the required symmetry are defined in that context. In a general spacetime, local and global problems are not in-

terchangeable. The theorem teaches precisely that conditions must be preserved, not forgotten once the slogan is memorized (Noether, 1918/1971).

Symmetry may exist in the rules without existing in the chosen state. Balance an idealized pencil on its tip in a perfectly symmetric arrangement. Surrounding directions are not favored by the description, yet the fallen pencil will have a particular direction. The example is physically imperfect but shows the logical difference: the state can select something the law does not select.

This is one route by which new phenomena arise without rewriting fundamental rules. If many components reach an ordered collective state, the useful description may include variables not obvious when looking at a single component. Symmetry breaking forces us to distinguish law, state and observable response (Burgess, 2007).

We can now formulate a question about mind. Why do we like a theory's symmetry so much? Because it reduces the number of cases needing separate treatment and reveals what is structural rather than accidental. But this preference is epistemically justified only when the structure produces testable results. An elegant equation can organize a wrong idea as elegantly as a good one.

In a thought experiment, a researcher discovers that their instrument always gives the same result after rotation. They might conclude that the world is isotropic, without a privileged direction at that level. But the instrument might instead be insensitive to the difference sought. Invariance of observation is not automatically invariance of reality. We must know which variations the instrument could have detected.

This problem will return in the simulation hypothesis. If a program computed space on a lattice, certain implementations might introduce preferred directions. If we do not find them, it does not follow that every simulation is excluded; it follows that we have not found the signature that this model class promises in the tested domain.

Symmetry is thus both a unifying power and a lesson in precision. It does not merely say the world is beautiful. It says which changes should not matter within a particular framework and sometimes offers an exact way to be wrong.

08. The arrow of time and order born of probability

A broken glass does not spontaneously reassemble on the table. Perfume released into a room does not ordinarily gather back into its bottle. In our experience, time has a direction. How does this asymmetry arise if some microscopic descriptions allow reversible evolution?

Before the physical answer, we can understand a counting idea. Consider twenty ideal, independent coins with equally probable faces. There is a single configuration in which all show the same chosen face. There are 184,756 configurations with exactly ten faces of one type and ten of the other. The calculation is the binomial coefficient ‘20 choose 10’. We have not conducted a study: this is a mathematical result for the specified model.

If an observer records only how many coins show each face, very many distinct configurations become the same coarse state for them. Here lies the idea of microstate and macrostate. The microstate preserves details; the macrostate groups situations agreeing in the quantities tracked. A macroscopic state may be much more frequent not because each realization is favored, but because it has far more realizations.

This example does not by itself prove the second law of thermodynamics. Coins are not molecules, and the choice of distribution and dynamics matters. It does explain, however, why order at the distribution level does not require special instructions to every component to produce the collective result.

Boltzmann entropy links, within the appropriate framework, a macrostate to the logarithm of the number or volume of microstates compatible with it. Entropy is not simply ‘disorder’ in the sense of an untidy room. The notion depends on the physical description and relevant variables. An alphabetically ordered library has no special thermodynamic value for that reason alone.

There is another decisive condition: the past. Starting from a special state and examining typical evolutions compatible with it can explain the direction of macroscopic processes. But why did that special state exist? Discussions of the hypothesis of a low-entropy past show that dynamical laws and cosmological conditions are not the same piece of the explanation. Various philosophical programs try to integrate them differently (Chen, 2020).

This is a general lesson about the origin of observed laws. A regularity may depend on the rules of evolution plus the state from which the world begins. If we forget the second term, we attribute to dynamics a power it does not possess alone. Two universes with the same equations and different initial conditions need not have the same macroscopic history.

In small systems, fluctuations cannot always be treated as negligible details. Relations such as Jarzynski's equality link far-from-equilibrium processes to free-energy differences through precise statistical averages. This does not authorize miraculous machines; it refines how we relate thermodynamics to outcome distributions (Jarzynski, 1997).

Notice the change of perspective. A highly reliable law at everyday scale may have a statistical formulation at another level. This does not cancel its practical reality. Nor does it mean every improbability becomes a reasonable hope. 'It is not forbidden in every microscopic detail' and 'it is worth expecting to happen' are radically different claims.

Life does not refute entropy increase merely because organisms maintain ordered structures. An organism exchanges matter and energy with its environment; looking only at local order means conveniently cutting out the system. Analysis must track exchanges and balances, not merely compare two pictures, one of a cell and another of a mixture.

More philosophically important: if the arrow of time also depends on a special condition of the world, part of what we experience as everyday necessity carries the memory of a history. The universe can produce stable regularities not readable solely in its list of interactions. Sometimes visible law is the meeting of dynamics, statistics and the past.

09. How new rules arise without a new legislator

Imagine inhabitants of a universe containing only a large network of locally interacting elements. They discover persistent objects, waves, boundaries between regions and abrupt transitions. One declares: every new phenomenon proves that the world's author added a rule. Another asks: might the same rule, repeated over many components, produce all this?

The second possibility is real in mathematical models. Cellular automata update cell states through local rules. Some systems with extremely simple descriptions have rich behavior. Matthew Cook proved the computational universality of Rule 110 in suitable configurations. The result concerns a formal system and its ability to simulate computation; it does not prove that our universe is a cellular automaton or that an arbitrarily chosen initial state will generate civilizations (Cook, 2004).

It does not follow that every complexity comes from a short program. A two-line program replaying a gigantic file does not explain the file's information. Rule simplicity must be evaluated together with the simplicity of the initial state and boundary conditions.

Consider a more modest model: on an unbounded line, independent particles take equal steps left or right with equal probabilities. There is no individual instruction to 'spread out the concentration'. Yet the collective distribution has calculable properties.

If each step has length a , mean displacement is zero and mean squared displacement after n steps is $n \cdot a^2$. Independence and equal probability are premises, not discoveries about every particle. The calculation follows because cross-correlation terms between steps average to zero. The characteristic distance thus grows as the square root of the number of steps. We have obtained a collective regularity without a coordinator tracking the entire cloud.

In appropriate limits, such models lead to diffusion descriptions. The point is not to identify every real molecule with a game coin, but to see the explanation's form: specify the microscopic mechanism, choose collective variables, justify approximations and derive a relationship at another scale. 'Emergence' becomes informative only when we can show such transitions.

Sometimes multiple microscopic mechanisms lead to similar macroscopic behaviors. This is the idea of universality in precise contexts. Details differing between systems may become unimportant for certain quantities at certain scales. The renormalization group provides a framework for studying how descriptions transform when scale changes and for separating relevant details from those that matter increasingly less (Burgess, 2007).

Coarse-graining represents multiple fine states at a common level; it does not physically erase details. Sometimes their effects are suppressed in the observables tracked; at other times ignored information returns and the model fails. The justification depends on the system.

This relative autonomy of levels explains why we can do science without knowing the ultimate basis. An inhabitant of a simulated world could discover correct fluid laws without knowing the processor. We can describe a material through collective variables without tracking each constituent. Being effective does not mean being fictional. It means being true or useful at a specified resolution and under specified conditions.

At least two meanings of emergence must nevertheless be distinguished. In the weak sense, collective behavior depends on components and rules but is difficult to anticipate or requires a new description. In the strong sense, properties or powers arise that would not, in principle, be determined or explainable by the complete basis according to the adopted criterion. The second meaning is a far more demanding thesis. Complexity and our surprise alone do not prove it.

Saying water 'invents' liquidity can be a good metaphor but does not replace analysis. First we must establish what we call a property, what 'derivable' means and what information about the basis we have al-

lowed. Sometimes we cannot calculate. Sometimes we lack suitable concepts. At other times we claim ontological independence. These problems must not be crammed into one word.

Can new regularities arise without new fundamental rules? Yes, and models show us how. Emergence does not automatically explain why basic rules, an initial state or something rather than nothing exists. Demanding all these answers would turn explanation into a magic word.

10. More than the sum of particles

Saying that a novel consists of characters is true. Saying that understanding the alphabet suffices to explain betrayal in the novel is absurd. Yet the fictional characters require no additional letters outside the text. The example shows that dependence on a substrate and explanatory sufficiency of the substrate's vocabulary are not the same thing.

'Reductionism' designates several theses. One is ontological: everything in the domain discussed depends on physical components and their relations. Another is methodological: to understand, we break the system into parts and investigate mechanisms. A third is explanatory: all good explanations can be replaced without significant loss by explanations at the lowest level. We can accept the second's usefulness without having proved the first, and accept the first without being committed to the third.

A molecular explanation of a substance may be indispensable for one question and needlessly detailed for another. If I want to know why a bridge collapsed, I may need material properties, geometry, loads and maintenance history. Even an available complete list of particles does not automatically tell me which comparisons matter for design and responsibility.

Think of a program bug. It is physically realized by circuits and memory. But the explanation 'this branch does not handle the zero case' may identify the problem better than a description of every electrical switching event. We invoke no supernatural power of software. We choose a level at which regularity and intervention are intelligible.

The same principle helps us discuss top-down causation without mysticism. A channel's shape constrains water flow. The channel's geometry is realized through matter; it does not come from outside physics. Yet to explain the difference between two flows, the relevant vari-

able may be shape rather than the complete microscopic inventory. Some debates on emergent causation pursue a stronger thesis, but the ordinary example of constraints does not prove it alone.

The critic of reductionism is right to demand attention to organization, context and levels. They are wrong to think ‘the whole’ explains something before specifying through which relations it does so. A sentence such as ‘the system has a holistic energy’ does not become an explanation merely because it sounds more generous than the vocabulary of mechanisms. It must show effects, conditions and observable differences.

There is also a mathematical argument against an overly naive reduction: knowing microscopic rules does not guarantee a general algorithm answering all macroscopic questions. Gu and colleagues construct models in which certain properties of infinite systems are undecidable. The result has technical premises and does not automatically transfer to every finite material, but it destroys the simple inference ‘known rules, therefore every property can be computed’ (Gu et al., 2009).

It does not follow that the higher level introduces new matter. It follows that ontological reduction, if true, is not the same as having a computational method or short explanation available. We may live in a world entirely dependent on a few rules and still need different disciplines to understand it.

Rigorous pluralism does not put all explanations on an equal footing. Biological explanation must be compatible with relevant physical constraints; economic explanation cannot ignore resources; psychological explanation cannot turn every adverse result into confirmation. But compatibility does not require eliminating the vocabulary of functions, information, institutions and motives.

The useful question is not ‘are you a reductionist or a holist?’ as though choosing a camp. It is: for this problem, which details are necessary, which relationships are robust and which intervention would distinguish the explanations? Sometimes we must descend to components. At other times we must raise our gaze until we see the configuration that makes them relevant.

A world of simple rules is not necessarily a world with simple explanations for every question. This may be the most important correction to the dream of a ‘theory of everything’: even if we obtained an extraordinary fundamental theory, we would still have to discover what happens when everything is put together.

11. Life: regularities with memory

If an organism survives, has it obeyed the ‘law of the strongest’? Only if we retrospectively call every surviving organism ‘strong’. But then we have explained nothing. We have renamed the result.

In evolutionary theory, the relevant success is not reducible to muscular strength or individual survival. What matters is differential transmission of traits across generations in a concrete environment. An advantage in one environment can become a cost in another. Selection can favor resilience, fecundity, cooperation, conflict avoidance or exploitation of others, depending on relationships and conditions. It needs no conscious goal and guarantees no moral progress (Darwin, 1859/1872).

This changes the meaning of a biological law. Part of the explanation is general: variation, heredity, reproductive differences. Another part is historical: which variations existed, in what order they appeared, which environment they encountered and what other traits accompanied them. The general principle does not replace a case’s causal history.

We can see the difference in an elementary genetic model. If two variants, A and a , have frequencies p and q , with $p + q = 1$, and the union of the two genetic contributions is independent and random, pair probabilities are p^2 , $2pq$ and q^2 . The calculation is the expansion of $(p + q)^2$. To interpret it as a Hardy–Weinberg equilibrium, biological assumptions must also be specified, such as the absence of forces altering frequencies in the idealized model. The conditional equation is useful precisely because deviations force us to seek which condition is unmet.

Biology is no less scientific for giving conditions importance. The problem would be presenting conditions as optional details. No prediction about a trait’s success is serious if it forgets the environment in which success is measured.

There is another complication: organisms modify the environment selecting them. A human community can change its institutions, resource access and support for children or vulnerable people. Without attributing precise genetic mechanisms to particular social changes, we can observe the logical relation: if environment matters for advantage, changing the environment can change the advantage. ‘Natural’ does not mean ‘independent of what we do’.

Darwin also discusses social instincts, sympathy and sexual selection. His writings nevertheless contain judgments marked by his era’s hierarchies and prejudices. Reading them as foundational texts does not require turning every passage into contemporary authority. The principle of selection must be separated from speculative historical applications and moral hierarchies attributed to human groups (Darwin, 1871/1874).

When someone invokes ‘the laws of biology’ to declare that social assistance contradicts nature, the answer need not be denying evolution. We ask which model, variable and evidence support the claim. Support, cooperation and institutions are also processes in the natural world. The existence of selection does not entail that we ought to let someone suffer, just as the existence of infections does not entail that we ought to avoid treatment.

Who benefits biologically from the ability to identify regularities? In an environment where anticipation changes opportunities for action, a system learning useful relationships may have an advantage. But ‘useful’ and ‘completely true’ are not identical. A simplified rule can guide local action well and mislead in another domain. This is a possibility argument, not proof that every human error is a selected adaptation.

This is precisely where improvised evolutionary stories must be avoided. We can quickly invent a story about the advantage of a belief, generosity or cruelty. Empirical support requires comparisons, mechanisms, measurements and alternatives. That an explanation seems plausible retrospectively does not prove the process happened that way.

Nor do economic and biological success overlap by definition. Income, status, number of children, survival of descendants and well-being are different variables. Saying a group ‘wins’ without specifying the indicator may conceal a change of subject. Having more descendants does not demonstrate intellectual or moral superiority; having more money does not demonstrate a hereditary advantage.

Life ultimately teaches us to conceive laws without erasing history. General regularity and contingent event are not adversaries. A rule can explain why certain differences matter, while history can explain why precisely those differences were available. Between them appears a world where success is real, contextual and in no way a cosmic verdict on a being’s worth.

12. Can laws change?

If we melt ice, the world after melting seems to have different rules. A rigid object has become a liquid. We can attribute the change to temperature and collective state without assuming fundamental interactions were rewritten. This is the first, common and important sense in which ‘new laws appear’: the domain and effective description change.

The second sense concerns parameters. A relationship may retain the same form, but a quantity we considered constant may depend on time, location or an unknown field. In a broader theory, what was a fixed parameter can become a dynamical variable. We have not eliminated the idea of law; we have moved the boundary between rule and state.

Searches concerning the constancy of fundamental quantities are genuine physical research. Atomic-clock comparisons, spectral observations and other systems allow variations to be tested with different sensitivities and systematic errors. Uzan surveys both methods and the need for coherent theoretical frameworks to interpret results. This literature does not identify variation with proof of simulation (Uzan, 2011).

One technical detail has enormous philosophical importance: we measure ratios. If we say a dimensional constant ‘changed its value’, we must specify units and standards. A unit change can change the written number without changing the world. Dimensionless quantities, such as certain ratios and dimensionless constants, avoid this direct ambiguity (Uzan, 2007).

Return to the two imaginary clocks. If both accelerated by exactly the same ratio, together with every process through which we observe and remember, we would have no unchanged internal standard against which to detect the acceleration. If one changes its rate relative to the other, there is a measurable difference. An experiment has no access to ‘time outside the universe’ merely because we call an instrument a clock.

The third sense of change would be more radical: the fundamental rule itself would change. But we can immediately ask whether the modification follows another rule. If so, we have a metalaw. If not, we must explain what ‘follows no rule’ means and how we would distinguish this from a rule we do not yet know.

A universe in which parameter k increases according to $k(t) = k_0 + at$ is not a lawless universe. It has a richer rule than $k = \text{constant}$. A universe in which k is modified by an external operator would have uncontrolled inputs from its internal perspective. But the operator’s behavior might itself be regular, random, goal-dependent or entirely opaque to observers.

Thus, ‘laws evolve’ is not yet an explanation. We must specify what evolves, relative to which time, through what mechanism and with what consequences. Otherwise we move the mystery from one noun to another. An explanation of change presupposing a possibility space, update rules and a probability measure already has a lawlike structure it must acknowledge.

There are also hypotheses attempting to explain why we observe particular parameter values. In a hypothetical ensemble of different regions or universes, observers would appear only under conditions compatible with their existence. This observational selection may matter but does not itself produce the ensemble or establish how probable its variants are. Saying ‘we had to observe a world in which we can exist’ is less than explaining that world’s mechanism.

Lee Smolin proposed cosmological natural selection: in a speculative scenario, black holes give birth to descendant universes with parameter variations. Universes producing more black holes would have more descendants. The analogy tries to replace arbitrary choice of constants with a mechanism of variation and reproduction. It is not a confirmed observation of universes being born or proof that life is the cosmos’s purpose (Smolin, 2006).

The dispute shows why precision matters even in speculative research. In a 2006 critique, Vilenkin argues that changing the cosmological constant could increase black-hole production, contradicting the invoked optimization criterion. Smolin responds by challenging the physical assumptions used in that argument. We cannot turn this con-

troversy into an experimental verdict on universe reproduction. We can see, however, that a hypothesis gains content when it produces a constraint and critics can say exactly where it would fail (Vilenkin, 2006; Smolin, 2006).

In such scenarios, effective laws can have a genealogy. But the mechanism of generation, variation and selection itself presupposes rules. The problem is displaced and may become more precise without disappearing. A mechanism for the emergence of constants would be a major discovery even if it did not explain why the possibility space itself exists. Partial explanation is not useless merely because the ultimate question remains open.

We can also conceive a situation with no simple ultimate level. The world might allow successive descriptions without an available final foundation. This is a metaphysical possibility, not an established conclusion of physics. Likewise, a very simple fundamental structure may exist while we cannot prove there is no deeper one.

What would living with this uncertainty mean? Not nervously checking gravity at every step. Practical confidence must relate to evidence and the domains where it works. But neither should we declare an anomaly impossible merely because it would contradict the textbook. Textbooks are documents about knowledge, not contracts signed by the universe.

A law can be revised for three very different reasons: we measured incorrectly; we discovered the domain was narrower; we found different dynamics. Only the last could support certain senses of profound physical change, and even it does not automatically identify an external author. When we say 'nature changed its rules', we must be sure we are not merely describing the moment we understood them better.

13. The brain that demands explanations

You hear a noise in an empty house. The mind immediately seeks something that produced it. In many situations this reflex is useful. But ‘what produced it?’ can turn too quickly into ‘who wanted to produce it?’ Between detecting a relationship and attributing an intention lies a difference that language easily hides.

Psychology offers no single verdict on why people believe in laws of nature. Research studies more bounded processes: pattern identification, explanations of properties, responses to uncertainty and how people generalize. Applying these to the metaphysics of laws is an interpretation that must be announced, not a direct experimental result.

Cimpian and Salomon propose the ‘inherence heuristic’: a tendency to explain some patterns through supposedly inherent characteristics of things without sufficiently examining history and external conditions. It is a theoretical framework supported by several research lines, not a universal law of judgment. A social pattern may seem natural because we attribute its cause to a group’s nature, although the conventions producing it should also be investigated (Cimpian și Salomon, 2014).

The connection to our topic is clear, but it is an extrapolation: ‘that is how things are’ can stop exploration of mechanisms. If a child asks why a social role is more often occupied by a particular group, ‘because those people are made for it’ sounds explanatory. In reality, it may close precisely the investigation of access, selection, learning and expectations.

Whitson and Galinsky reported, across six experiments, a connection between lack of control and illusory pattern perception in different tasks. The result became attractive for the idea that the mind compensates for uncertainty with order. But this story must not be presented as definitive fact (Whitson și Galinsky, 2008).

Van Elk and Lodder reported seven conceptual-replication experiments in which they did not find expected effects on the measures used. Their analyses questioned the robustness of control manipulations and indicated publication-bias problems in the examined literature. This does not prove that the need for control never matters. It shows that an elegant psychological explanation can be more fragile than its popularity (Van Elk și Lodder, 2018).

It is a perfect lesson for a book about laws: we cannot combat naive belief in regularities by ourselves using an insufficiently verified psychological regularity. The psychology of science must undergo the same discipline it demands of science.

There is also a deeper logical error. Suppose someone believes in a law because they need security. This psychological origin does not imply the law is false. A person may believe something true for a weak reason and something false for an intelligent reason. A belief's genealogy and testing its content are different investigations.

How might the effect of the word 'law' be studied more directly? Here is a proposal, not an experiment already conducted: participants receive the same numerical regularity labeled either 'law', 'provisional model' or 'observed pattern'. They are then asked how far they generalize, which counterexamples they accept and which interventions they consider possible. Wording, numerical competence and source prestige should be controlled, and hypotheses and analysis established before seeing results.

An interesting result would be if the label 'law' changed confidence even when data and precision were identical. Absence of an effect would be equally important. It should be repeated across languages, contexts and populations before we speak of a general mechanism. A student study cannot directly establish the psychology of all societies.

Education could train two complementary reflexes: courage to seek regularities and patience to put them to the test. Telling students 'think critically' is not enough. We must give them cases where a seductive explanation and a correct one make different predictions, then let them observe the difference.

The mind needs compression: we cannot preserve every experience as an episode unrelated to the rest. The danger is not the existence of mental patterns. It is forgetting the distinction between a reassuring pattern and one that survived a serious attempt to contradict it.

14. What does understanding actually mean?

One person may correctly calculate an object's orbit without intuitively imagining it in every detail. Another may draw a highly convincing but incorrect picture. Which one understands? The question shows that we use the same word for different competencies.

We can recognize a pattern, predict, explain a mechanism, identify what would change if we intervened and connect the result to a more general structure. These capacities sometimes support one another but do not coincide. A results table can predict without explaining. A causal story can seem clear and fail numerically.

Understanding also has an aspect of transfer. If I memorize an answer for one situation without knowing when it applies, my competence is fragile. If I can systematically modify conditions and anticipate which part of the answer remains, I have grasped something deeper. A well-understood law is not merely a correctly reproduced formula, but a map of dependencies and limits.

Mathematics extends this ability because it allows precise tracking of relationships, including when visual intuition gives way. But it does not eliminate the representation problem. We must decide which quantities correspond to the world, what we ignore and which measurements can check the result. An impeccable calculation may answer an unsuitable question.

Why does the world seem so accessible to mathematical descriptions? We can consider their success evidence of its structure. We can also notice selection: we call precisely the descriptions that worked 'discovered laws', while failed attempts disappear from textbooks. The two observations do not cancel each other. Selection explains part of our retrospective picture without making the existence of exploitable regularities trivial.

Nor is there a simple reason why an evolved brain had to understand the entire universe. For many biological functions, distinguishing opportunities and dangers at the scale of its life would suffice. Microscopic or cosmological understanding is achieved through cultural extensions: languages, instruments, mathematics, institutions of verification. This observation about supports' role establishes no fixed limit to mind; it shows that 'what humans can do' is not reducible to an unaided individual's intuition.

Relativity offers a historical example: understanding required revising assumptions about simultaneity, clocks and geometry, not merely more precise measurements in the old vocabulary. In his exposition for the public, Einstein connects concepts to measurement procedures and shows why time cannot simply be treated as an identical background for every observer. In general relativity, spacetime geometry participates in describing gravity. These conceptual changes do not mean nature changed its rules when physicists changed the theory (Einstein, 1916/1920).

Our ability to learn such a framework matters for the question of mental limits. An intuition formed in everyday circumstances is not an infallible judge of physical possibility. But not every counterintuitive idea deserves adoption either. The gain comes from testable relationships enabled by the new framework, not its difficulty or distance from common sense.

Artificial intelligence makes the difference between finding a relationship and human understanding of it more visible. The AI Feynman system uses symbolic regression and physics-inspired heuristics to recover formulas from data, evaluated partly on a set of known equations. The results show the power of guided-search methods; they do not demonstrate automatic discovery of the universe's foundations (Udrescu și Tegmark, 2020).

SINDy seeks dynamical descriptions with few nonzero terms chosen from a candidate library. It makes a fruitful assumption explicit: dynamics may be simple in suitable coordinates and bases. Success nevertheless depends on representation, data and the list of possibilities supplied to the method. Finding the equation in a chosen vocabulary is not identical to inventing the relevant vocabulary (Brunton, Proctor și Kutz, 2016).

Now imagine an AI predicting an experiment better than humans but producing a model nobody can explain in one page. There are at least three options. We can reject the model because it does not resemble our understanding. We can trust it without limit because it predicts well. Or we can separate use from justification: test it outside its construction data, investigate stability and seek partial explanations of important mechanisms.

The third option is most promising. It does not demand that every truth fit our working memory, but neither does it transfer authority to an opaque box. We can understand conditions under which a result is reliable even without holding every operation that produced it in mind. At the same time, for risky interventions, prediction without knowledge of limits may be insufficient.

Can there be a true law the human mind cannot understand? It depends on ‘understand’. Perhaps we cannot visualize it. Perhaps we cannot compute its consequences unaided. Perhaps we cannot formulate suitable concepts. These difficulties do not prove a definitive cognitive frontier exists. But past success does not guarantee that every structure of the world will become transparent to us either.

The most productive question is more concrete: what extension would let us understand the next relationship? A notation, a new experiment, a simulation, a verification tool or a collective concept? Sometimes the mind does not overcome a limit through greater effort in the same form. It changes its way of working.

15. Are there things even intelligence cannot compute?

‘If we knew all the laws, we could predict everything.’ This sentence compresses at least three assumptions: that we also knew the relevant state, that we could perform the computation and that the question had a solution accessible through a general method. None comes free with the list of equations.

The first limit is observational. We measure with a certain precision and do not control every condition. In a sensitive system, very small initial differences can grow over time until detailed predictions diverge. Lorenz’s model shows how simple deterministic equations can produce nonperiodic evolutions sensitive to initial state. Determinism is not equivalent to practical predictability (Lorenz, 1963).

This does not mean we can say nothing. Some statistical properties or outcome ranges may be predictable even when the exact trajectory is not. Moving from ‘I do not know the detail’ to ‘everything is equally probable’ is an error. The prediction limit must be formulated at the level of the question and time horizon.

The second limit concerns resources. A computation possible in principle may require too much time, memory or energy for a concrete observer. Seth Lloyd’s research on physical limits to computation links computational capacity to resources and physical theory. These are not proofs about every metaphysically possible substrate; they are limits formulated within specified physical frameworks (Lloyd, 2000).

The third limit is more severe: absence of a general algorithm for a precise problem class. Cubitt, Pérez-García and Wolf proved undecidability of the spectral-gap problem for certain families of quantum lattice systems in the thermodynamic limit. This does not mean every finite material has a property impossible to investigate. It means the desired universal procedure solving every instance of the constructed class does not exist (Cubitt, Pérez-García și Wolf, 2022).

It is tempting to use such theorems as a general certificate of mystery. That would be a mistake. An undecidability proof has a domain and premises. It does not tell us the particular problem on our desk is insoluble, that intuition solves it miraculously or that science has reached its end. It forces us to relinquish a universal promise, not the investigation of cases.

Nor do Gödel's theorems authorize 'every important truth exceeds reason'. They concern limits of formal systems with precise properties. Leaping from arithmetic to claims about soul, cosmos or an AI's omniscience requires additional arguments. Invoking a mathematical name is not one of them (Raatikainen, 2025).

There is another limit, often more relevant than computational difficulty: identifiability. Can two different mechanisms produce exactly the same available observations? If so, an observer cannot separate them merely by analyzing the same data more carefully. This is not lack of intelligence but lack of a difference in accessible information.

An elementary example: a thermometer displays 20, and we do not know whether the temperature is 20 and the instrument correct, or the temperature is 18 and the instrument has a constant two-unit error. More identical records do not resolve the ambiguity. An independent standard or another intervention type may resolve it. The problem requires not merely more computation, but different information.

In the simulation hypothesis, this distinction becomes decisive. A simulated and a nonsimulated world might have the same results for every permitted internal experiment. In that case, lack of evidence would not necessarily stem from a simulator's conspiracy. It would stem from observational equivalence of the descriptions.

There is also selection among the questions we can formulate. An observer built from a world's processes can access only certain variables. But we should not turn this observation into a grand paradox before describing a model. We can prove limits in a well-defined formal system; we cannot assume every imagined limit applies to our universe.

Being rigorous about limits paradoxically means being more ambitious. Instead of saying 'it cannot be understood', we ask: can it not be measured, not be computed with available resources, does no general

algorithm exist, or is there no observable difference? Each diagnosis suggests a different research program. Confusing them produces needless resignation and impossible promises.

16. Faith, meaning and alternatives to mechanism

When someone says the universe cannot be ‘just a machine’, they may express very different objections. They may demand an explanation for consciousness, defend freedom, assert a creator’s existence or reject the idea that a life’s value is exhausted by a physical description. We should not answer all of them with the same argument.

Theism can conceive order as dependent on a God who creates and sustains the world. Deism, schematically, emphasizes a creator and the autonomous operation of the created order. Other positions identify the divine with nature or attribute a more complex relationship to it. These orientations do not in themselves make the same prediction about experiments. A religious conception can be compatible with the same regularities accepted by a naturalist.

The differences may be metaphysical and existential without an immediate measurable consequence. But if a doctrine asserts that a particular physical event occurs under specified conditions, that component can be confronted with observation. Shielding an empirical claim with a general conception’s prestige is an error, whether that prestige is religious or scientific.

The problem of miracles shows the tension. If we define a miracle as violating a true and absolutely universal law, we have a contradiction by definition. If we define it as an unexpected event relative to known laws, it remains open whether we have an error, unknown mechanism or intervention. Hume analyzes justification for belief in miracles by weighing testimony against experience of regularities. Even those rejecting his conclusions can retain the methodological question: why should this account weigh more than alternative explanations? (Hume, 1748, sec. X.)

An advocate of a created world can legitimately ask why order exists. A naturalist may reply that introducing a creator moves the question: why does that creator exist and have those powers? Nor does ‘the laws

simply exist' eliminate explanation's stopping point. The dispute also concerns what kind of endpoint is acceptable, not merely whether current knowledge has a gap.

Panpsychism, broadly, assigns fundamental status to experiential aspects or precursors of experience. It does not necessarily imply that stones think like humans or that our wishes alter instruments. It may try to answer a question about consciousness's nature while remaining compatible with the same physical predictions. But that compatibility is not experimental confirmation of the thesis (Goff, Seager și Allen-Hermanson, 2022).

Holism may demand that relationships and organization not be eliminated from explanation. This demand is sometimes indispensable. It becomes weak when it refuses to specify what difference the whole would make and how it could be investigated. 'Everything is connected' says too little: how, how strongly, at what scale and with what consequences?

Another alternative is process philosophy: what we call objects can be understood through the stability of processes rather than as substances wholly independent of change. We need not declare this orientation the final truth to use its question. A whirlpool persists through matter's movement; an institution persists through activities and rules. When stability is produced, explaining it requires more than identifying a substrate (Seibt, 2022).

In *Beyond Good and Evil*, §22, Nietzsche attacks how 'nature's lawfulness' is presented as an uninterpreted text of the world. His challenge is that the concepts through which we read nature may carry preferences and values. We need not accept every consequence of his philosophy to hear the warning: even the ideal of a completely regular world may have a cultural history (Nietzsche, 1886).

But this genealogy does not make every interpretation equal either. Two vocabularies may be culturally laden, yet one may enable far better predictions and interventions. Acknowledging interpretation does not mean abandoning reality's resistance. The strongest critique preserves both levels.

Instead of automatic contempt or admiration, we ask what problem the alternative solves, what additional assumptions it makes and what would count against it. Some ideas will remain metaphysical; others will suggest experiments or disappear when precision is demanded.

Perhaps the world has a meaning physics does not formulate. Perhaps meaning is something beings construct in the world. Neither possibility entitles us to falsify an experimental result. Respect for great questions begins by not using answers too large for evidence too small.

17. Who speaks in nature's name

'It is not my decision. That is how the world works.' Few formulas can move responsibility so quickly. Sometimes the speaker correctly warns us of a constraint. At other times they try to turn a contestable choice into an inevitability.

There is no single class of people benefiting from the concept of a law of nature. A reproducible result can protect a vulnerable person from arbitrariness and give a powerful organization a new instrument of control. The same physical principle can make a device safer or a weapon more effective. The distribution of benefits depends on access, aims and institutions, not merely the law's truth.

The first gain is reduced dependence on the authority's person. If we can show how a result is obtained and under what conditions, competence can be transmitted. A student need not inherit a teacher's prestige to verify a calculation. This is the ideal. Realizing it nevertheless requires instruments, time, education and freedom to contradict, all unequally distributed.

The sociology of science shows why declared universality of criteria does not automatically eliminate hierarchies of recognition. Merton analyzed the 'Matthew effect': contributions associated with already prestigious researchers may receive disproportionate credit, while similar contributions from lesser-known researchers remain less visible. His analysis concerns recognition mechanisms, not proof that scientific results are merely products of reputation (Merton, 1968).

The distinction is crucial. The social origin of trust does not automatically establish the truth or falsity of its object. It may explain why a correct idea is ignored or a weak one receives more attention. Evaluation also requires content, replication and independent results.

Nosek, Spies and Motyl discuss the tension between incentives for publishable results and incentives for accuracy. Favoring novelty and positive results can affect analytical choices and the visibility of negative findings. Their proposals concern research practices and rewards.

Here we see that trust in laws is supported not only by intelligent minds but also by institutions making correction advantageous (Nosek, Spies și Motyl, 2012).

The second gain is anticipatory power. Whoever knows a regularity before others can design, prevent or exploit. This asymmetry is not inherently illegitimate: competence is valuable. It becomes problematic when affected people cannot see assumptions, contest errors or negotiate consequences.

Imagine a model estimating a person's probability of dropping out of a training program. The organization uses it to deny access to those deemed 'risks'. The denial may prevent precisely the experience that would have changed the outcome. Prediction becomes part of the mechanism. We no longer merely observe a relationship: we use it to modify the world in which it was measured. The example is hypothetical, but shows why a social regularity cannot automatically be treated as a planetary orbit.

The third gain is legitimacy. When a hierarchy is called natural, its opponent seems to fight reality rather than an institution. We must decompose the claim: there is an observed difference; there is an explanation of causes; there is a judgment about what should be done. The first alone does not yield the second, and the first two do not automatically yield the third.

Saying a result has a biological component does not establish what moral treatment the person deserves. Saying a social measure has costs does not establish that it is unjustified. Saying an ideal is good does not prove the proposed mechanism will realize it. Rigor demands that facts, causes and values remain distinct, then enter explicit dialogue.

Who loses when the notion is misused? The person turned into a statistical average, the community told change is impossible and the researcher whose results do not fit the dominant story. Who loses when we reject the notion altogether? Those who depend on verifiable standards for safety, fair access and challenging arbitrariness.

The alternative is not between obedience to experts and distrust of all expertise. It is an organization of trust in which competence has authority within its domain while decision-making power remains ac-

countable for values, distribution and errors. Nature does not vote on the budget or decide who deserves to suffer. Humans make these choices, sometimes using knowledge of nature, sometimes hiding behind it.

18. Profit, merit and the supposed laws of the market

A company cannot produce material goods without resources and energy. But it does not follow that a particular profit distribution is a physical law. Between material constraint and economic outcome lie rights, contracts, technologies, information and institutional choices.

An economic model can be highly useful without the status of a universal law. It can show what consequences follow from particular preferences, rules and constraints. The problem arises when premises disappear from the presentation and the conclusion is promoted to humanity's eternal nature.

Construct a simple example. Two people divide ten resource units. If one receives seven, three remain for the other. This is an arithmetic constraint. But arithmetic did not decide who receives seven. There may be a contract, an auction, an urgent need, a prior contribution or a threat. Invoking the fixed sum does not justify the distribution rule.

Moreover, the total resource can sometimes be changed through cooperation, investment or technology, and sometimes cannot within the relevant horizon. Thus, the opposition between 'everything is zero-sum' and 'we can always increase the total' is too crude. We need a model of the concrete constraint, costs and time, not optimism or pessimism turned into law.

The economic benefit of knowing physical laws can be analyzed without mythology: it allows anticipation of material responses, design and reduction of some errors. But a process's predictability does not determine who owns its result. Technological advantage and the right to exploit it are distinct levels. An invention may increase total well-being yet produce losses for particular people.

Suppose automation reduces the time needed for a product. No single social order follows from this change. Prices may fall, profits may rise, working hours may decrease or different combinations may emerge.

Competitive conditions, negotiation and rules matter. Saying ‘technology decided’ removes from the story the agents who chose the effects’ distribution.

Some regularities change precisely because people react to them. If an institution rewards the number of documents produced, people may produce more documents without improving the activity for which documents were an indicator. In our hypothetical model, the target changed behavior; the old correlation between indicator and quality is no longer guaranteed. This is a general reason to test relationship stability after intervention, not a claim that every evaluation is useless.

Merit is equally subtle. Success can include competence and effort, but these are not the only possible explanations of a difference. Initial access, the order of encounters and others’ reactions can alter the trajectory. Before proclaiming a ‘law of talent’, we must ask which alternatives were eliminated and how talent was measured.

Salganik, Dodds and Watts’s cultural experiment involved over 14,000 participants who could listen to and download unfamiliar songs. Knowing others’ previous choices increased inequality and unpredictability of success across experimental worlds. Quality, operationalized through participants’ independent choices, did not become irrelevant; but it did not determine the ranking alone. The result has the limits of an artificial market and is not a complete theory of merit (Salganik, Dodds & Watts, 2006).

The implication for our discussion is that an outcome can be produced by stable rules yet not reflect a unique, inevitable hierarchy. The same interaction types can amplify early differences. We need not choose between ‘everything is merit’ and ‘everything is luck’. We must identify contributions, interactions and possibilities for change.

Who benefits from the rhetoric of a perfectly natural market? An actor wanting their advantage no longer discussed may benefit. Who benefits from denying all economic regularities? An actor promising results without acknowledging required resources and incentives may benefit. Both can be ways of evading responsibility.

Mature economic language says: under these conditions, this mechanism tends to produce that result; if conditions change, we must re-evaluate the prediction. This is not a weakness. It is the difference between a tool for choice and an oracle claiming choices have already been made by nature.

19. Art between constraint and invention

A violin cannot produce every imaginable sound through any gesture. A sculptural bridge must bear its own weight. A painter works with light, pigments, surfaces and a visual system. Art is not a territory outside nature. But these constraints do not yield a single legitimate style or an eternal canon of beauty.

Three levels must be separated here: physical possibility, perceptual effect and cultural value. Physics can help us understand sound production. Psychology can investigate what particular people distinguish or prefer under particular conditions. Art criticism and history discuss meaning and value in context. No level can automatically replace another.

A numerical ratio between frequencies is a measurable property. The claim that the ratio suits every music, listener and purpose is much stronger. Even if we found a robust perceptual preference, it would not follow that art ought to maximize it. The artist may pursue unease, ambiguity, tension or surprise, not merely comfort.

The same happens with visual symmetry. It can be an organizing tool. Asymmetry can express movement or conflict. Our ability to recognize a pattern does not oblige every work to reproduce it. A constraint is a condition of the game; an aesthetic norm is a choice about how it is worth playing.

Claims about universal proportions supposedly explaining beauty on their own also deserve caution. Retrospectively finding a ratio in a sufficiently complex object is easy if we can freely choose measurement points. A test would require measurements fixed beforehand, comparisons with alternatives and an effect criterion. Without these, mathematics can ornament an impression rather than explain it.

The concept of law also influences art through order's prestige. An artist may seek the world's essence in geometric forms, repetition or rule variations. Another may think precisely the rule's rupture makes

experience visible. We need not choose either intention as nature's truth. We can judge what the work accomplishes and what question it forces us to formulate.

Generative art offers an especially clear conceptual laboratory. We fix a few rules and obtain a family of results. From this family we select some works. What part of the value comes from the rule, what part from the initial state and what part from selection? If we show only successful results, the audience may attribute to the algorithm a power the complete process does not possess to the same extent.

This is not an accusation against algorithmic generation. Artists working manually also select, revise and abandon. The question is what we present as the creative mechanism. 'A simple rule produced this image' can be true and incomplete if thousands of failed attempts preceded the choice. The full explanation also includes the filter.

This observation returns us to emergence. A complex form may arise from simple rules, but not every complexity is meaningful to an observer. Meaning may depend on experience, conventions and intentions. Explaining the pixel distribution does not mean explaining why someone sees loss or promise in it.

Who benefits when an aesthetic norm is called 'natural'? Those who master it and institutions using it as a filter may gain legitimacy. Who loses? Forms of expression evaluated by a criterion concealing its contextual character. But pluralism does not require asserting every object is equally well made either. We can assess coherence, difficulty, precision and effect without pretending our preference is written in the atom's structure.

Perhaps the most interesting link between art and laws of nature is not that art illustrates them. It is that art teaches us to notice how many worlds of meaning can be built without changing basic constraints. The same sounds can belong to different songs. The same materials can become shelter or monument. Freedom does not necessarily begin where rules disappear; sometimes it begins where we understand what a rule permits without commanding.

20. The simulation hypothesis without cinema

If we learned the universe is simulated, what exactly would we have learned? That matter has a different substrate from the one imagined? That authors exist? That they can intervene? That the world's purpose is an experiment? These conclusions are often packaged together although the first does not prove all the others.

First, three ideas must be separated. The world can be described computationally. The world can have fundamental dynamics resembling computation. The world can be implemented in another system as a simulation made for particular purposes. Using computers to describe gravity does not prove gravity is produced by an external computer.

Nor is opposition between emergence and simulation mandatory. Emergent effective laws can arise within a simulation. A nonsimulated system could have simple discrete rules. A simulator could implement continuous or probabilistic processes through methods unlike our video games. The two questions differ: how do a level's regularities arise, and on what substrate is that level realized?

Nick Bostrom's argument is more precise than the popular formula 'we probably live in a game'. In a conditional scheme, it links three possibilities: civilizations like ours rarely reach the relevant capability; those reaching it produce very few ancestor simulations; or simulated observers of the relevant type greatly outnumber nonsimulated ones. Moving to a belief about ourselves nevertheless requires assumptions about consciousness, resources, civilizations' behavior and the observer class from which we consider ourselves a sample (Bostrom, 2003).

It does not follow that the three alternatives have equal probabilities. A disjunction containing three statements does not allocate one third of reality to each. Nor does technical ability to simulate a process

prove that the simulated process has conscious experience. These difficulties are part of evaluating the argument, not details to leave until after the conclusion.

We can construct an arithmetic example without claiming it describes the cosmos. If there are a thousand nonsimulated observers and a million simulated observers of a comparable type, a uniform sample from their union will almost always be simulated. But why should our situation be equivalent to that uniform sample? How do we define 'comparable'? Do we count people, moments of experience, identical copies or worlds? The arithmetic is simple; justifying the sampling scheme is not.

Another vocabulary error appears. A simulation need not be 'unreal' to its inhabitants. Chalmers argues that a Matrix-type hypothesis can change the world's metaphysics without making all ordinary judgments about its objects and relations false. If an object has stability and causal roles in our world, discovering a computational substrate may change what we believe it fundamentally is, not the fact that we can bump into it (Chalmers, 2003/2005).

Nevertheless, simulating an environment for an external brain must be distinguished from minds themselves implemented within a simulation. The first presupposes a mind on another substrate; the second presupposes that simulated organization can sustain experience. Arguments about consciousness cannot be resolved simply by counting characters generated in a program.

Intervention is not mandatory either. A program can be left to evolve, receive inputs or be modified. Its creator may have complete control, limited control or no longer exist. Deducing 'constantly supervised by an omnipotent intelligence' from 'implemented' imports a theology without argument.

What do we gain intellectually from the hypothesis? It forces us to distinguish internal law from implementation mechanism, prediction from ontology and experience from substrate. What do we risk? Turning every surprise into a message and every absence of evidence into evidence of simulation's perfection. In the latter case, the hypothesis becomes immune to experience.

The serious question is therefore not ‘can it be imagined?’ It can. The question is which precise version we choose, what it adds to an internal physical description and which observations should differ. Without this step, we have a possible metaphysics. With it, some variants can become research programs.

21. What would count as evidence?

Two observers find a perfect square in a particle distribution. The first says ‘the simulation grid’; the second, ‘the structure of space’. The observation is the same. What must be separated is not two exclamations, but models with different predictions.

This is the central difficulty. ‘We are in a simulation’ designates an enormous family of possibilities, not a unique mechanism. A simulation may approximate some processes or reproduce them exactly at the level accessible to observers. It may have a regular lattice, an adaptive representation or an implementation unlike a video game. That we use particular architectures does not oblige every possible simulator to use them.

Let us accept the generous condition that the simulator does not actively try to hide the implementation. Even then detection is not guaranteed. Lack of camouflage does not automatically produce an information leak. Two different mechanisms can have the same observable consequences, just as two programs can compute exactly the same function. If every accessible experiment has the same outcome distributions in the two models, no statistical ingenuity can create an absent difference. This is an identification limit, not a limit of the researcher’s courage.

An example of a more precise hypothesis was explored by Beane, Davoudi and Savage. They analyzed consequences of implementation on a cubic spacetime lattice under a particular Wilson-type discretization scheme. Such a construction can privilege certain directions and affect very energetic particles. The paper discusses constraints and possible signatures of that model; it offers no universal test for every simulated universe (Beane, Davoudi și Savage, 2014).

The distinction is decisive. If we found anisotropy, a difference between directions, we would have a physical structure to explain. It could come from the proposed implementation model, a nonsimulated fundamental structure, particle sources, propagation or instru-

ments. A positive result would not arrive labeled ‘external author’. A negative result would constrain certain implementations, not exclude every imaginable simulation.

Nor are quantum phenomena, by themselves, evidence that the world is computed only when we look. In physics, measurement involves physical interactions, not necessarily a human mind. Decoherence describes the loss of local access to interference through a system’s interaction with the environment and explains important aspects of classical behavior’s emergence. It does not alone resolve every dispute over quantum mechanics’ interpretation. But neither does it require a conscious spectator activating reality’s ‘rendering’ (Zurek, 2003).

Indeed, a test attempting to discover whether the world computes details only on demand faces a problem: the very device intended to record the detail becomes a relevant interaction. We must specify what ‘on demand’ means in the simulation model and what different outcome distribution it produces. The analogy with saving resources in a game is not yet a physical prediction.

We could seek rounding errors, generator periodicity or resolution limits. But every proposal requires a sufficiently precise architecture. A finite sequence cannot alone prove that its source is fundamentally random; it can be reproduced by a program containing it. Conversely, unexpected regularity in a sequence does not prove a programmer’s existence. It may indicate an ununderstood physical process, data selection or an instrument defect.

More promising, at least logically, would be an anomaly that not only surprises but anticipates. An implementation model could predict a family of correlated deviations in advance: a particular energy dependence, orientation and ratio between quantities. If these relationships appeared in independent data, the model would gain support. It would still need comparison with physical explanations producing the same family of effects.

There is also the message scenario. Suppose a repeatable signal appears that responds to new challenges through controlled physical channels. After ordinary explanations were excluded, it would be an extraordinary discovery of a source capable of intelligent interaction.

But ‘an intelligence communicates’ and ‘this intelligence is outside our universe’ are different conclusions. External origin must be supported separately, not extracted from spectacular content.

A project can be valuable without finding a simulation: it improves clocks, characterizes noise or tests symmetries. A weak project changes its explanation after every result and leaves nothing verifiable. Research productivity matters, not the amazement produced.

Some implementations might leave traces; others would remain undetectable. Research begins when the proposed mechanism precisely indicates what we should observe differently.

22. A protocol for a suspicious world

Return to the prologue's clocks. The ratio between two frequencies starts changing. One team announces an error. Another, an unknown physical field. A third, intervention in the simulation. How do we organize research so that more than the story with the best headline can win?

First we specify the phenomenon. Not 'time runs differently', but a particular dimensionless ratio between two transition types, measured through an identifiable protocol, deviates from a model by a certain magnitude and temporal structure. Choosing the ratio prevents a mere unit change from being confused with changed physics. Research on varying constants uses precisely comparisons where physical dependencies and model assumptions are explicit (Uzan, 2011).

Then we formulate sufficiently concrete competing explanations. One model describes thermal drift in the instrument. Another, a synchronization-chain problem. A third introduces a field affecting the two mechanisms differently. A fourth proposes a discrete modification of an implementation rule. The list need not be complete; we must not treat our preferred model as the sole alternative to standard theory.

For each explanation we ask what else should change. Does the effect appear in every instrument or only one batch? Does it depend on temperature, position, material or software version? Does it have the same form in laboratories sharing neither instruments nor code? Independence means more than multiple addresses. Ten teams using the same faulty library can perfectly reproduce the same mistake.

Predictions must be recorded before choosing the data they will evaluate. If we examine thousands of frequencies, periods and directions and report only the unusual combination, chance can acquire the appearance of law. We separate exploration from confirmation: find a pattern in one observation set, then put it at risk on new data. We also

retain negative results. Such practices address a general research problem: incentives for publishable results do not automatically coincide with incentives for true results (Nosek, Spies și Motyl, 2012).

The intervention model must also pay the price of precision. ‘Someone changes anything, anytime, however they want’ offers no determinate probability of observations. A table of exceptions invented after the result can reproduce any history. By contrast, a hypothesis specifying a particular class of modifications and predicting as-yet unmeasured relationships can be compared with rivals. It need not claim to explain everything to deserve testing.

We can build a small research program without waiting for a cosmic anomaly. The project proposed here would begin with artificial worlds whose mechanisms are known only to the evaluator. We do not claim this experiment has been conducted. Its purpose would be to test whether our methods distinguish possible causes of a change before applying them to the universe.

In the first family, rules remain fixed but the observer sees only collective variables. In the second, a regime change brings other effective relationships into view. In the third, an internal parameter evolves under a stable rule. In the fourth, the evaluator modifies a rule externally. In the fifth, the world does not change at all: the sensor or processing software fails. The observer receives the same interface types and a comparable experimental budget.

The difficulty must be constructed correctly. If every external intervention is abrupt and every internal process smooth, an algorithm will learn the distinction we artificially chose. Internal jumps, slow external interventions and errors resembling both must be included. The mechanism’s true label must not be guessable from a data-generator artifact.

Evaluation occurs on new worlds, not merely new rows from the same worlds. Otherwise the method may memorize a simulator’s peculiarities and seem to discover general principles. We compare the detector with simple methods and measure false alarms, detection delay and the proportion of cases where it justifiably refuses to decide. Saying ‘these two explanations cannot be separated by the available data’ may be an epistemic success, not a failure.

We deliberately introduce pairs of observationally equivalent worlds as well. In one, a hidden variable changes the dynamics; in the other, the evaluator externally applies exactly the same change. If interfaces produce the same observations under every permitted intervention, no detector should reliably distinguish them. A system announcing certainty in such a situation has not overcome the theoretical limit. Most likely it exploited an information leak or evaluation error.

This laboratory would produce useful results even if it showed how little we can identify. We would learn which new experiments reduce ambiguity, which conclusions depend on hidden-variable access and when classification must be replaced with a set of compatible explanations. In an AI program for scientific discovery, the ability to declare underdetermination should be evaluated alongside the ability to find formulas.

For real observations, the next step would be a public register of predictions, protocols and data, with independent checks and explicit stopping rules. A campaign's result must be formulated at the level it supports: 'this model class is incompatible with the data in this domain', 'there is an unreproduced deviation', or 'the new model predicts these observations better'. None of these statements should automatically become 'we found the programmer'.

There is also a mundane but important ethical limit: no simulation hypothesis justifies causing disasters to force the system to 'react'. Experiments must meet the same safety requirements as other research. A method claiming to test reality by sacrificing people already starts from a moral conclusion the data cannot authorize.

The program is useful because it shifts emphasis from fascination to discrimination. The question is no longer whether we can tell a plausible story after seeing a change. It is whether we can anticipate which observation would distinguish two mechanisms and honestly acknowledge cases where no accessible observation distinguishes them.

23. If the rules are rewritten tomorrow

In a novel, an author can change the world's rule from one page to the next. An inhabitant of the novel has no access to the manuscript. For them, the question is not only what changed but whether change can still be described through something stable. This is a thought experiment about access to explanation, not proof that the world has an author.

At least three intervention types must be separated. Someone can modify the system's state without modifying evolution rules: move a particle, then let it evolve. They can modify a parameter that seemed constant in the internal description. Or they can replace the rule linking the present state to the next. The three operations have different consequences, and observers may struggle to identify them from a finite outcome sequence.

Take a simple game. A ball abruptly changes position, then continues under the same dynamics. This resembles a state intervention. If all balls begin responding differently to the same force, we suspect a parameter or mechanism change. If, under certain circumstances, the dependency itself takes another form, we suspect a rule change. But a richer model could describe all these phenomena through fixed rules and hidden variables.

Thus, 'a fundamental law changed' is more ambitious than 'our constant-parameter model failed'. The second conclusion can be experimentally supported. The first requires knowing that no more general level exists where the change follows a stable rule. Finite experience does not usually provide such a guarantee.

But what if the simulator stops everything for an hour and then resumes? If precisely all accessible processes are suspended, including clocks, memory and every external interaction, that external hour leaves no internal trace. We would not feel a wait, because feeling requires an unfolding. The situation would change if one channel con-

tinued functioning and could be compared with suspended ones. Detectability depends on a difference preserved in observations, not a duration imagined from outside.

The same argument applies to uniform slowing of the entire simulated world relative to implementation time. Internally, there is not necessarily an independent clock to measure the slowdown. 'The universe runs more slowly' may designate a relationship between two levels without designating an observable change at ours. By contrast, a slowdown affecting processes differently could leave measurable ratios.

Could we exclude simulation through its computational cost? Only conditionally. Physical limits to computation link processing to resources such as energy, time and degrees of freedom within the assumed physics (Lloyd, 2000). Transferring these limits to an external simulator requires justifying why its implementation would obey the same physics and possess the assumed resources. Invoking an external universe with whatever capability we want does not prove simulation; but neither does an internal limit automatically exclude every imaginable external world.

Then comes the metalaw problem. If our world's rules are changed by a program, the program operates under other rules. If those are changed by an intelligence, that intelligence and its acts presuppose some mechanism or structure of possibilities. Explanation through simulation can move 'why these laws?' to another level. It need not resolve it all the way down.

Nor does an omnipotent-author hypothesis eliminate every question. We can ask what its power means, whether it acts for reasons, whether those reasons offer predictions and why such a capability exists. Some theological answers consider this a foundation no longer requiring the same kind of explanation. Other philosophies reject the stopping point. We have no simple experiment arbitrating among all these forms of ultimate explanation.

The practical consequence is not abandoning knowledge. We already live using models with domains and conditions. If we discover a change, we seek what remains stable: more general relationships, pre-

dictable intervals, verification procedures. A world in which some regularities change is not identical to one in which every statement has the same value.

The moral consequence requires still more attention. If simulated people have conscious experiences, being implemented does not make their suffering nonexistent. This is an ethical conditional: it does not claim every program is conscious. It says that where experience exists, we cannot morally erase it by renaming its substrate. A creator's power to stop a system does not, by that fact alone, confer a moral right to do anything to its inhabitants.

Who would benefit from the contrary interpretation? An authority able to declare others' lives mere dispensable elements of a scenario. Who would lose? Precisely those whose experience is ignored. The risk is not unique to simulation: the same operation occurs whenever a metaphysics is used to diminish beings' moral status without an argument about their interests and experiences.

In a world with uncertain ontology, we can prefer robust decisions: reducing avoidable suffering, preserving the ability to correct mistakes, protecting research and avoiding catastrophic wagers on fragile hypotheses. These priorities are arguable normative proposals, not new laws of nature. Precisely for that reason, they must be owned as responsible choices.

If tomorrow we discover that rules can be rewritten, the difference between truth and invention does not disappear. What we must discover changes. And the most urgent question may not be 'who owns the code?' but 'which relationships can we understand well enough to protect one another?'

24. How to live without the final explanation

We can now answer the opening question without pretending to close it. Are laws of nature real or imagined? Statements, equations and classifications are human constructions. In many tested situations, the constraints they try to describe do not seem to depend on their formulator's desire. The existence of additional metaphysical necessity remains disputed. These three answers can be held simultaneously because they answer three different questions.

We need not choose between two caricatures: a universe with a perfect legal code on its walls and a mind arbitrarily inventing every order. There is a world that resists and beings that select, compress and interpret that resistance. Knowing means making selections that remain useful when the world has an opportunity to contradict us.

What did we gain by calling stable relationships 'laws'? The ability to coordinate experience across generations, build instruments and demand more from an explanation than the speaker's prestige. What did we risk? Confusing an approximation with destiny, a local regularity with universal necessity and a power relationship with existence's inevitable order.

This dual power is no reason to abandon the concept. It is a reason to use it with an identification sheet. What does it describe? Under what conditions? With what precision? What observation would challenge it? Who can verify? Who bears the error? The last two questions do not replace physics, but prevent its use as an excuse for unacknowledged allocation of risks.

On emergence, the conclusion is also dual. Stable basic rules can produce new collective relationships and indispensable descriptive levels. It does not follow that every novelty requires breaking physics. But neither does a list of components automatically provide a useful explanation of the whole. Sometimes we must discover suitable variables before recognizing the law.

On mind, it would be wrong to confuse lack of an intuitive image with lack of understanding. We can understand through relationships, consequences and limits, supported by mathematics, instruments and cooperation. It would be equally wrong to confuse a prediction's success with possession of the final explanation. Mature understanding also knows what it has not obtained.

On simulation, the fruitful question remains conditional. Some implementations might leave traces; others might be observationally indistinct from nonsimulated explanations. Absence of intent to camouflage does not negate the second possibility. Research begins when we choose a precise mechanism class and explain what differences it could produce, not when we declare every anomaly confirms our story.

On morality, neither natural selection, the market nor a cosmic program decides by itself what we owe others. A description of mechanisms helps choose means, anticipate effects and avoid impossible promises. Aims require arguments about values, interests and relationships. Refusing to confuse the two does not weaken science; it asks that science not be used in place of deliberation it cannot replace.

My position is that we need more realism about constraints and less certainty about their ultimate interpretation. Realism prevents us from treating every desire as a solution. Caution prevents us from treating every temporary solution as the universe's final ruling. Between them lies room for intellectual courage, not merely modesty.

Let us return one last time to the imaginary laboratory. The clocks still disagree. After months of work, perhaps we discover a trivial error. Perhaps we find a physical phenomenon. Perhaps the relationship between observations and explanations changes in a way we cannot yet name. In every variant, the good researcher has gained something: more precise knowledge of what happens and what they are not yet entitled to claim.

The world need not read equations for some equations to describe it well. We must carefully read the conditions under which we use them. And when someone says 'nature decided', perhaps the most human question is this: what exactly have we discovered about the world, and what exactly are we trying to stop discussing among ourselves?

A. Four tools for not confusing story with law

1. Simplicity of the rule is not simplicity of the world

A complete computational explanation must specify at least the rule, initial state and how the result relates to observation. A program of a few characters displaying a huge file does not explain that file through a few characters: the information has been stored elsewhere. This elementary example shows why ‘can a few rules produce the universe?’ must be accompanied by ‘what have we already put into the initial conditions?’

We can compare two explanations of the same sequence. The first says ‘read the next entry in this table’, and the table contains all results. The second offers a short rule and short initial state from which results can be generated. Both reproduce the past; the second compresses information more. Yet past compression alone does not guarantee future prediction. We need testing on data unused in choosing the rule.

The same caution applies to emergence. If we already hide in the generator the complex structure we wish to explain, we have not explained its appearance. If a robust structure appears across many initial conditions and a class of mechanisms, we have better reason to speak of an emergent regularity. Robustness must be measured, not suggested by a spectacular animation.

2. A provable limit to simulation detection

Let S denote a precise simulated-world model and N a precise competing model we call nonsimulated. D is an observable event and E an experiment accessible to the observer. Assume the models give the same observation distribution: for every permitted experiment and every event D , they assign the same probability:

$$P(D \mid S, E) = P(D \mid N, E).$$

For any event with nonzero probability, the likelihood ratio is then 1. Those data favor neither model over the other. If we begin with different preferences, they may remain different, but this experiment has not made them better supported.

Adaptive experimental choice does not help either if equality holds after every possible observation history. The probability of an outcome sequence is obtained by multiplying successive conditional probabilities. If every factor agrees between models, so does the product. The adaptive strategy therefore produces the same distribution of observable histories. This is an elementary proof of indistinguishability under the stated conditions, not a claim that every simulation is undetectable.

The way out is not a miraculous statistical method. We must find an experiment for which equality's premises no longer hold, acquire a new channel or restrict the model class. Creative research can begin here. The clear limit shows where the difference must be sought.

3. Why a spectacular result does not eliminate alternatives

Invent two models, A and B. Before a test, assign A a probability of 1% and B the remaining 99%. Assume, strictly for this example, no other possibilities exist. An outcome D is ten times more probable under A than B. The initial A:B odds were 1:99. After the outcome they become 10:99, so A's probability reaches approximately 9.2%, not certainty.

The calculation does not establish what probability should be assigned to simulation. The numbers are chosen only to show the difference between 'the data favor a model' and 'the model is now almost certain'. A real problem requires justifying both observation probabilities under models and treatment of alternatives. A vague model that does not say how probable D is cannot receive a numerical advantage merely because the outcome can be narrated in its vocabulary.

If the observer searched hundreds of possible outcomes and chose the one favoring A, even the ten-to-one ratio cannot be accepted without examining selection. Data must be analyzed alongside the procedure by which they were sought and selected. Arithmetic precision does not repair a poorly defined comparison.

4. The profile of a purported law

A useful profile begins with the exact statement and domain: which objects, scales and conditions? It continues with observable quantities, units and measurement procedure. It includes error tolerance, auxiliary assumptions and the difference between an exact relationship in the model and an approximation in application. These few details are worth more than an unexplained adjective ‘universal’.

The second part records support: what was measured, derived or merely assumed? Which new prediction succeeded? Which laboratory or independent method verified it? What result would seriously weaken confidence? When such a result appears, we explain why we revise the law, conditions or instrument; we do not automatically choose whichever option preserves the author’s authority.

The third part examines transfer and social power. Was the relationship tested in the context where it is now used? Is it a physical constraint or an institution-dependent regularity? Who gains resources, prestige or impunity through this formulation? Who bears an error’s cost? These questions do not decide whether the law is true, but clarify why it deserves independent testing.

The final field separates descriptive from normative conclusions. ‘Under conditions C, behavior X produces outcome Y’ does not mean ‘we must preserve C’ or ‘Y is good’. A law may show how an effect is obtained; choosing the effect and allocating costs require additional arguments. This is the minimal discipline by which law remains a tool of knowledge rather than a command to stop thinking.

B. Glossary: words that change the question

Empirical adequacy. A theory's ability to correctly represent relevant observable phenomena. It is not by definition identical to the complete truth of its ontology.

Anisotropy. Dependence of a property on direction. An observed anisotropy requires explaining source, propagation and measurement before attributing it to a fundamental structure.

Causality. The relationship through which some changes matter for producing others. Philosophical definitions and identification methods differ; mere correlation does not alone fix causal direction.

Initial condition. Description of the state from which we apply an evolution rule. Information placed here must not be confused with information explained by the rule.

Dimensionless constant. A quantity whose numerical value does not depend on unit choice. It is particularly important when asking whether physical relationships, rather than measurement conventions, have changed.

Counterfactual. A statement about what would happen in a situation different from the actual one. Laws are often used for such questions, not merely to summarize the past.

Decoherence. A process through which interaction with the environment makes certain interference locally inaccessible. It does not in itself presuppose a conscious observing mind.

Determinism. A model property in which the relevant state and rules determine a unique evolution. It is not synonymous with practical predictability.

Domain of validity. The set of conditions under which a relationship can be used with the stated precision. An approximation can be excellent within its domain and inadequate outside it.

Weak emergence. The appearance of collective behaviors from basic interactions without postulating additional fundamental powers. Novelty of description does not imply violation of the mechanism.

Strong emergence. A family of positions holding that a whole introduces properties or powers irreducible in an ontological sense. This is a stronger thesis than computational difficulty.

Epistemology. The study of knowledge, justification and their limits. ‘How can we know?’ is not identical to ‘what exists?’

Deterministic chaos. Behavior in which sensitivity to initial conditions can severely limit predictions although the rule introduces no fundamental randomness.

Identifiability. The ability to distinguish models or parameters through accessible observations and interventions. More data of the same type do not resolve every lack of identifiability.

Induction. The passage from observed cases to more general or future expectations. Its practical usefulness is not a deductive guarantee of nature’s uniformity.

Effective law. A relationship formulated for particular scales and relevant variables that need not describe every detail of the underlying mechanism.

Statistical law. A relationship about distributions, frequencies or collective values. A rare individual outcome can be compatible with the law; specified probabilities matter.

Metalaw. A proposed rule for how other rules are established or modified. Explanation at this level does not automatically close the question of foundations.

Model. A selective representation constructed for a purpose. It can combine laws, approximations, conventions and assumptions about conditions.

Nomological necessity. What is necessary relative to a world’s laws. It must not be confused with logical necessity, where the alternative would produce a contradiction.

Ontology. Inquiry into what exists and existence's fundamental categories. The same predictions can be compatible with different ontologies.

Explanatory reductionism. The idea that one level's explanation can be obtained from a more basic level's description. It differs from the thesis that everything has a physical substrate.

Symmetry. A transformation leaving a model's relevant structure unchanged. It is not reducible to a drawing's visual symmetry.

State / rule. The state describes the system's situation; the rule specifies relationships or transformations. Changing state does not automatically change the rule.

Underdetermination. A situation in which available evidence is compatible with multiple explanations. It may be temporary or persist for every accessible observation.

Universality. In collective-system studies, the appearance of common behaviors in different microscopic mechanisms under specified conditions. It does not mean every detail is irrelevant.

Likelihood. The probability of data under a specified model. It is not the same as the model's probability after observing the data.

Bibliography

References are ordered alphabetically. In-text citations and the links below are clickable. Double years distinguish the original text from the specified edition or translation. Article links may lead to the authors' version or the journal record. Online sources were consulted in September 2026.

Aristotel. *Fizica*, book II. English translation by R. P. Hardie and R. K. Gaye. Internet Classics Archive, MIT.

Text of book II

Armstrong, D. M. (1983). What Is a Law of Nature? Cambridge University Press.

Bibliographic details and book description

Beane, S. R., Davoudi, Z. and Savage, M. J. (2014). Constraints on the Universe as a Numerical Simulation. *The European Physical Journal A*, 50, 148. DOI: 10.1140/epja/i2014-14148-0.

Preprint: arXiv:1210.1847

Bostrom, N. (2003). Are You Living in a Computer Simulation? *The Philosophical Quarterly*, 53(211), 243–255.

Author's text

Brunton, S. L., Proctor, J. L. and Kutz, J. N. (2016). Discovering Governing Equations from Data by Sparse Identification of Nonlinear Dynamical Systems. *Proceedings of the National Academy of Sciences*, 113(15), 3932–3937. DOI: 10.1073/pnas.1517384113.

Preprint: arXiv:1509.03580

Burgess, C. P. (2007). Introduction to Effective Field Theory. *Annual Review of Nuclear and Particle Science*, 57, 329–362. DOI: 10.1146/annurev.nucl.56.080805.140508.

Preprint: arXiv:hep-th/0701053

Carroll, J. W. (2020). Laws of Nature. Stanford Encyclopedia of Philosophy; substantive revision of 16 November 2020.

Reference article

Cartwright, N. (1999). The Dappled World: A Study of the Boundaries of Science. Cambridge University Press.

Bibliographic details and book description

Chakravartty, A. (2017). Scientific Realism. Stanford Encyclopedia of Philosophy.

Reference article

Chalmers, D. J. (2003/2005). The Matrix as Metaphysics. Text from 2003, included in C. Grau (ed.), *Philosophers Explore the Matrix*. Oxford University Press, 2005.

Author's text

Chen, E. K. (2020). From Time Asymmetry to Quantum Entanglement: The Humean Unification. Preprint arXiv:2006.05029.

Text and versions

Cimpian, A. and Salomon, E. (2014). The Inherence Heuristic: An Intuitive Means of Making Sense of the World, and a Potential Precursor to Psychological Essentialism. *Behavioral and Brain Sciences*, 37(5).

Article at publisher

Cook, M. (2004). Universality in Elementary Cellular Automata. *Complex Systems*, 15(1), 1–40. DOI: 10.25088/ComplexSystems.15.1.1.

Article at journal

Cubitt, T. S., Pérez-García, D. and Wolf, M. M. (2022). Undecidability of the Spectral Gap. *Forum of Mathematics, Pi*, 10, e14. DOI: 10.1017/fmp.2021.15. Extended version of the result, initially released as a preprint in 2015.

Preprint: arXiv:1502.04573

Darwin, C. (1859/1872). *On the Origin of Species*. First edition: 1859; sixth edition: 1872.

Text: Project Gutenberg, no. 2009

Darwin, C. (1871/1874). The Descent of Man, and Selection in Relation to Sex. First edition: 1871; second edition: 1874.

Text: Project Gutenberg, no. 2300

Descartes, R. (1644). Principia Philosophiae / Principles of Philosophy. For laws of motion: part II. The indicated electronic edition contains selections in John Veitch’s translation; for Cartesian physics’ context, see also Slowik.

Selections: Project Gutenberg, no. 4391

Einstein, A. (1916/1920). Relativity: The Special and General Theory. English translation by R. W. Lawson, Methuen, 1920; the author’s preface is dated December 1916.

Text: Project Gutenberg, no. 5001

Goff, P., Seager, W. and Allen-Hermanson, S. (2022). Panpsychism. Stanford Encyclopedia of Philosophy; substantive revision of 13 May 2022.

Reference article

Griffel, F. Al-Ghazali. Stanford Encyclopedia of Philosophy. Used for the interpretive debate about necessity, occasionalism and causality.

Reference article

Gu, M., Weedbrook, C., Perales, Á. and Nielsen, M. A. (2009). More Really Is Different. Physica D: Nonlinear Phenomena, 238, 835–839. DOI: 10.1016/j.physd.2008.12.016.

Preprint: arXiv:0809.0151

Hume, D. (1748). An Enquiry Concerning Human Understanding. Sections IV–VII: induction and causality; section X: miracles. Electronic text based on later editions.

Text: Project Gutenberg, no. 9662

Jarzynski, C. (1997). Nonequilibrium Equality for Free Energy Differences. Physical Review Letters, 78, 2690–2693.

Preprint: arXiv:cond-mat/9610209

Kant, I. (1781/1787). Kritik der reinen Vernunft / The Critique of Pure Reason. First and second editions. English translation by J. M. D. Meiklejohn.

Text: Project Gutenberg, no. 4280

Kant, I. (1783). Prolegomena zu einer jeden künftigen Metaphysik / Prolegomena to Any Future Metaphysics. Especially §§14–38.

Text: Project Gutenberg, no. 52821

Laozi, traditional attribution. Dao De Jing / Tao Te Ching. English translation by James Legge (1891). Especially chapters 1, 25 and 37.

Text: Project Gutenberg, no. 216

Lloyd, S. (2000). Ultimate Physical Limits to Computation. Nature, 406, 1047–1054.

Preprint: arXiv:quant-ph/9908043

Lorenz, E. N. (1963). Deterministic Nonperiodic Flow. Journal of the Atmospheric Sciences, 20, 130–141.

Text hosted by the University of Vermont

Maudlin, T. (2007). The Metaphysics Within Physics. Oxford University Press.

Bibliographic details and book description

Merton, R. K. (1968). The Matthew Effect in Science. Science, 159(3810), 56–63. DOI: 10.1126/science.159.3810.56.

Text hosted by the University of Pennsylvania

Newton, I. (1687/1726). Philosophiae Naturalis Principia Mathematica. First edition: 1687; third edition: 1726. The indicated electronic text reproduces the third Latin edition.

Text: Project Gutenberg, no. 28233

Nietzsche, F. (1886). Jenseits von Gut und Böse / Beyond Good and Evil. Especially §22. English translation by Helen Zimmern.

Text: Project Gutenberg, no. 4363

Noether, E. (1918/1971). Invariante Variationsprobleme / Invariant Variation Problems. Translation by M. A. Tavel, *Transport Theory and Statistical Physics*, 1, 186–207 (1971). DOI: 10.1080/00411457108231446.

Text: arXiv:physics/0503066

Nosek, B. A., Spies, J. R. and Motyl, M. (2012). Scientific Utopia: II. Restructuring Incentives and Practices to Promote Truth over Publishability. *Perspectives on Psychological Science*, 7(6), 615–631.

Preprint: arXiv:1205.4251

Raatikainen, P. (2025). Gödel’s Incompleteness Theorems. *Stanford Encyclopedia of Philosophy*; substantive revision of 8 October 2025.

Reference article

Salganik, M. J., Dodds, P. S. and Watts, D. J. (2006). Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market. *Science*, 311(5762), 854–856. DOI: 10.1126/science.1121066.

Article and supplementary material

Seibt, J. (2022). Process Philosophy. *Stanford Encyclopedia of Philosophy*; substantive revision of 26 May 2022.

Reference article

Slowik, E. Descartes’ Physics. *Stanford Encyclopedia of Philosophy*. Used for the context of Cartesian mechanics, laws of nature and their relationship to metaphysics.

Reference article

Smolin, L. (2006). The Status of Cosmological Natural Selection. Preprint arXiv:hep-th/0612185. Theoretical proposal and defense, not a report of observed descendant universes.

Text and versions

Udrescu, S.-M. and Tegmark, M. (2020). AI Feynman: A Physics-Inspired Method for Symbolic Regression. *Science Advances*, 6, eaay2631. DOI: 10.1126/sciadv.aay2631.

Preprint: arXiv:1905.11481

Uzan, J.-P. (2007). Fundamental Constants. Preprint arXiv:0708.4361.

Text and versions

Uzan, J.-P. (2011). Varying Constants, Gravitation and Cosmology. *Living Reviews in Relativity*, 14, 2. DOI: 10.12942/lrr-2011-2.

Preprint: arXiv:1009.5514

van Elk, M. and Lodder, P. (2018). Experimental Manipulations of Personal Control Do Not Increase Illusory Pattern Perception. *Collabra: Psychology*, 4(1), 19. DOI: 10.1525/collabra.155.

Article at journal

Vilenkin, A. (2006). On Cosmic Natural Selection. Preprint arXiv:hep-th/0610051; revised version in November 2006. Critique discussed alongside Smolin's response.

Text and versions

Westerhoff, J. (2026). Nāgārjuna. *Stanford Encyclopedia of Philosophy*; substantive revision of 31 May 2026.

Reference article

Whitson, J. A. and Galinsky, A. D. (2008). Lacking Control Increases Illusory Pattern Perception. *Science*, 322(5898), 115–117. DOI: 10.1126/science.1159845. To be read alongside van Elk and Lodder (2018).

Article record

Zurek, W. H. (2003). Decoherence, Einselection, and the Quantum Origins of the Classical. *Reviews of Modern Physics*, 75, 715–775. DOI: 10.1103/RevModPhys.75.715.

Preprint: arXiv:quant-ph/0105127